CROP: Certifying Robust Policies for Reinforcement Learning through Functional Smoothing

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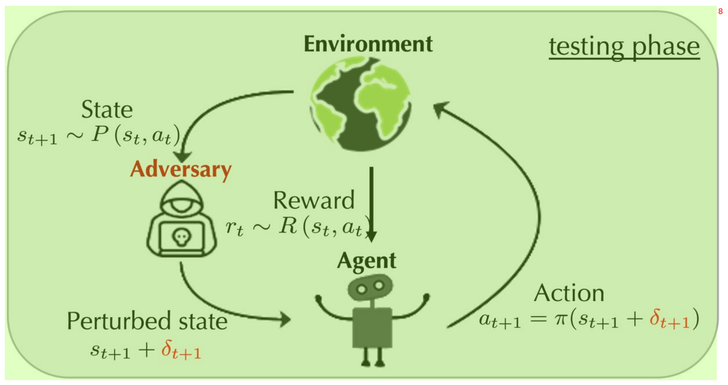
Background & Motivation

- Testing and improving the robustness of safety-critical applications before their massive deployment to real-world is of great importance.
- Robustness certification for a given learning algorithm is needed to end the repeated game between the attackers and defenders.

Threat Model

Testing-time evasion attack on agent's observation

Adversary can apply ℓ_2-bounded perturbations $\mathcal{B}^\varepsilon = \{\delta \in \mathbb{R}^n | \|\delta\|_2 \le \varepsilon\}$ to input state observations of the agent



Certification Goal

Provide test-time certification for the *performance* of the trained agent in the perturbed environment

1. Providing **criteria** for robustness certification
2. Proposing certification **strategies** corresponding to different certification criteria

Certification Criteria

Criterion 1 (state-level): Per-state action stability

- *Maximum Perturbation Magnitude* $\bar{\varepsilon}$:
$$\pi(s+\delta) = \pi(s), \forall \delta \in \mathcal{B}^{\bar{\varepsilon}}, \text{ where } \mathcal{B}^\varepsilon = \{\delta \in \mathbb{R}^n \mid \|\delta\|_2 \le \varepsilon\}$$

i.e., the policy produces the same action on state s with or without perturbation.

Criterion 2 (trajectory-level): Cumulative reward lower bound

- *perturbed cumulative reward* $J_\varepsilon(\pi)$
$$J_\varepsilon(\pi) := \sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(s_t + \delta_t))$$
where $s_{t+1} \sim P(s_t, \pi(s_t + \delta_t)), s_0 \sim d_0$
- *Lower bound of perturbed cumulative reward* $\underline{J}$: with perturbations at every step bounded by $\mathcal{B}^\varepsilon$, we have $\underline{J} \le J_\varepsilon(\pi)$.

CROP Framework

Per-State Action Certification

CROP-LoAct: Local randomized smoothing on states

Action-value functional smoothing

- Let $Q^\pi : \mathcal{S} \times \mathcal{A} \to [V_{\min}, V_{\max}]$, then

smoothed value function: $\widetilde{Q}^\pi(s,a) := \mathbb{E}_{\Delta \sim \mathcal{N}(0, \sigma^2 I_N)} Q^\pi(s + \Delta, a) \quad \forall s \in \mathcal{S}, a \in \mathcal{A}$

smoothed policy: $\tilde{\pi}(s_t) := \arg\max_a \widetilde{Q}^\pi(s_t, a) \quad \forall s_t \in \mathcal{S}$

Properties of the smoothed function

- Lipschitz continuity of $\widetilde{Q}^\pi$ w.r.t. the state input: $L = \dfrac{V_{\max} - V_{\min}}{\sigma}\sqrt{2/\pi}$
- lower bound of $\bar{\varepsilon}$:
$$r_t = \frac{\sigma}{2}\left(\Phi^{-1}\left(\frac{\widetilde{Q}^\pi(s_t, a_1) - V_{\min}}{V_{\max} - V_{\min}}\right) - \Phi^{-1}\left(\frac{\widetilde{Q}^\pi(s_t, a_2) - V_{\min}}{V_{\max} - V_{\min}}\right)\right)$$

Cumulative Reward Certification

CROP-GRe: Global smoothing on trajectories

Expectation bound via *mean smoothing*

- Mean smoothing: $\widetilde{F}_\pi\left(\oplus_{t=0}^{H-1}\delta_t\right) := \mathbb{E}_{\Delta \sim \mathcal{N}(0, \sigma^2 I_{H \times N})} F_\pi\left(\oplus_{t=0}^{H-1}(\delta_t + \Delta_t)\right)$
- Expectation lower bound:
$$\mathbb{E}\left[J_\varepsilon(\pi')\right] \ge \widetilde{F}_\pi\left(\oplus_{t=0}^{H-1}\mathbf{0}\right) - L\varepsilon\sqrt{H}, \text{ where } L = \frac{(J_{\max} - J_{\min})}{\sigma}\sqrt{2/\pi}$$

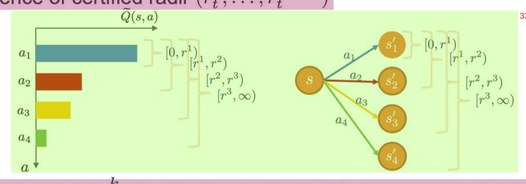
Percentile bound via *percentile smoothing*

- Percentile smoothing:
$$\widetilde{F}_\pi^p\left(\oplus_{t=0}^{H-1}\delta_t\right) = \sup_y \left\{y \in \mathbb{R} \mid \mathbb{P}\left[F_\pi\left(\oplus_{t=0}^{H-1}(\delta_t + \Delta_t)\right) \le y\right] \le p\right\}$$
- Percentile lower bound:

the p-th percentile of $J_\varepsilon(\pi') \ge \widetilde{F}_\pi^{p'}\left(\oplus_{t=0}^{H-1}\mathbf{0}\right)$, where $p' := \Phi(\Phi^{-1}(p) - \varepsilon\sqrt{H}/\sigma)$

CROP-LoRe: Local smoothing + adaptive search

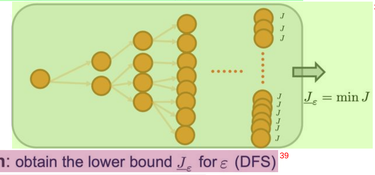
Sequence of certified radii $(r_t^1, \ldots, r_t^{|\mathcal{A}|-1})$



- r_t^k: when $\varepsilon < r_t^k$, the possible action will be among the set of top k actions
$$r_t^k = \frac{\sigma}{2}\left(\Phi^{-1}\left(\frac{\widetilde{Q}^\pi(s_t, a_1) - V_{\min}}{V_{\max} - V_{\min}}\right) - \Phi^{-1}\left(\frac{\widetilde{Q}^\pi(s_t, a_{k+1}) - V_{\min}}{V_{\max} - V_{\min}}\right)\right), \quad 1 \le k < |\mathcal{A}|$$

Adaptive search algorithm

- Base case: $\varepsilon = 0$, $\underline{J}$: reward without perturbation
- Adaptive search: alternating the two procedures:



1. Trajectory exploration and expansion: obtain the lower bound $\underline{J}_\varepsilon$ for ε (DFS)

2. Perturbation magnitude growth: obtain a new ε' (priority queue)

- Outcome of the algorithm: $\{(\varepsilon_i, \underline{J}_{\varepsilon_i})\}_{i=1}^{|C|}$

Experiments

Setup

- Four RL environments: Freeway, Pong, CartPole, Highway
- Nine DQN variants: StdTrain, GaussAug, AdvTrain, SA-MDP (PGD,CVX), RadialRL, CARRL, NoisyNet, GradDQN
- Two certification criteria: per state-action, cumulative reward
- Three certification algorithms: CROP-LoAct, CROP-GRe, CROP-LoRe

Robustness Certification for Per-State Action

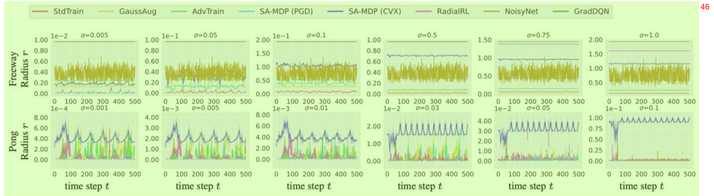


Fig. Robustness certification for per-state action in terms of certified radius r at all time steps.

- GradDQN and RadialRL achieve the highest certified radius on Freeway.
- Periodic patterns can be observed in Pong.
- Certified results largely match empirical observations.

Robustness Certification for Cumulative Reward

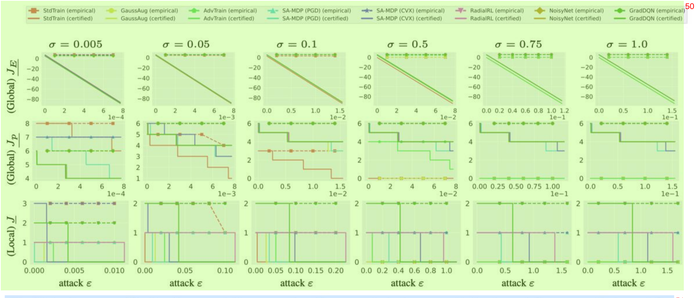


Fig. Robustness certification for cumulative reward on *Freeway*, including expectation bound $\underline{J}_E$, percentile bound $\underline{J}_p$, and absolute lower bound $\underline{J}$.

- Under global smoothing, percentile bound is tighter than expectation bound.
- For CROP-LoRe, different RL algorithms demonstrate different behavior in different regions of attack magnitude ε---GradDQN achieves higher lower bound under smaller ε, while RadialRL is the dominant one under larger ε.
- The certification is relatively tight, compared with the empirical result.

More details, experimental results, and code can be found in our paper, leaderboard, and repository!

Paper Leaderboard Repository